



Lecture 22.07: Differentiation Rules & Activation Function Derivatives

 Python 3.10+

 Numpy Numerical Computing

Status **Completed**

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1. Fundamental Rules of Differentiation

Rule Name	Mathematical Statement	Example
Power Rule	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[x^4] = 4x^3$
Constant Multiple Rule	$\frac{d}{dx}[c \cdot f(x)] = cf'(x)$	$\frac{d}{dx}[5x^3] = 15x^2$
Sum & Difference Rule	$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$	$\frac{d}{dx}[x^2 + 3x] = 2x + 3$
Product Rule	$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$	$\frac{d}{dx}[xe^x] = e^x + xe^x$
Quotient Rule	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$	$\frac{d}{dx}\left[\frac{x}{x+1}\right] = \frac{1}{(x+1)^2}$
Chain Rule	$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$	$\frac{d}{dx}[(3x+1)^4] = 4(3x+1)^3 \cdot 3$



2. AI Activation Functions & Their Derivatives

1. Sigmoid Function: $\sigma(x) = \frac{1}{1+e^{-x}}$

Using the Quotient / Chain Rule:

$$\sigma'(x) = \frac{d}{dx}[(1 + e^{-x})^{-1}] = -(1 + e^{-x})^{-2}(-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Rewrite in self-referential form:

$$\sigma'(x) = \frac{1}{1 + e^{-x}} \cdot \left(1 - \frac{1}{1 + e^{-x}}\right) = \sigma(x)(1 - \sigma(x))$$

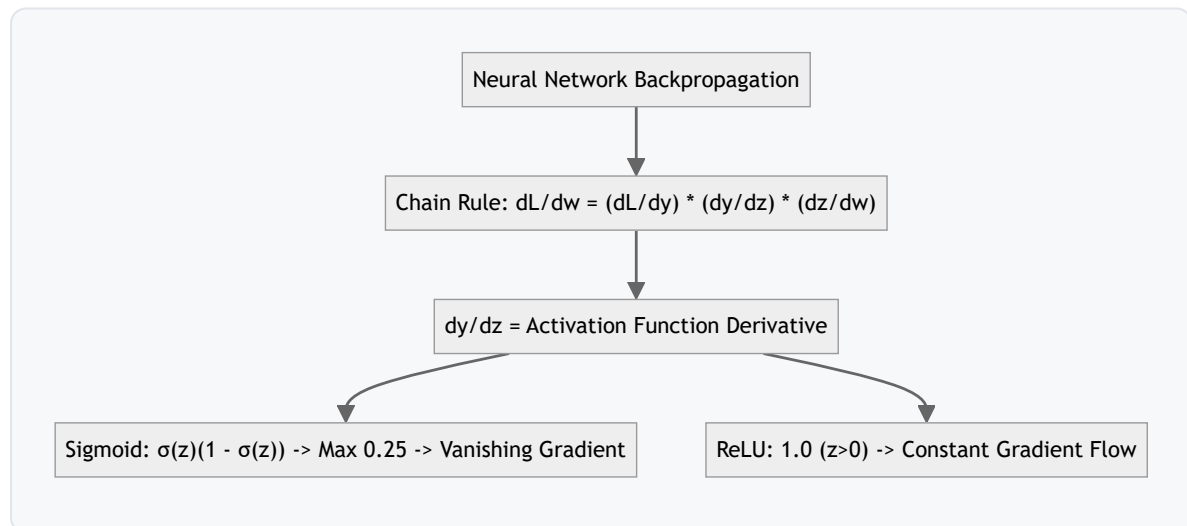
2. Hyperbolic Tangent: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$\tanh'(x) = 1 - \tanh^2(x)$$

3. Rectified Linear Unit (ReLU): $\text{ReLU}(x) = \max(0, x)$

$$\text{ReLU}'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

(At $x = 0$, the subgradient convention sets the derivative to 0 or 1).



3. Python Implementation: Verifying Activation Derivatives

```
import numpy as np

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def sigmoid_deriv(x):
    s = sigmoid(x)
    return s * (1 - s)

x_test = np.array([-2.0, 0.0, 2.0])
print("Sigmoid values:      ", np.round(sigmoid(x_test), 4))
print("Sigmoid derivatives:", np.round(sigmoid_deriv(x_test), 4))
print("Max Sigmoid derivative occurs at x=0:", sigmoid_deriv(0.0))
```